NOTING that it adopted, by resolution MEPC.328(76), the 2021 revised MARPOL Annex VI, which is expected to enter into force on 1 November 2022 upon its deemed acceptance on 1 May 2022,

NOTING IN PARTICULAR that the 2021 revised MARPOL Annex VI contains amendments concerning mandatory goal -based technical and operational measures to reduce carbon intensity of international shipping,

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SOURCE: MEPC.337(76).pdf
NOTING IN PARTICULAR that the 2021 revised MARPOL Annex VI contains amendments concerning mandatory goal -based technical and operational measures to reduce carbon intensity of international shipping,

NOTING FURTHER that regulation 28.4 of MARPOL Annex VI requires reference lines to be established for each ship type to which regulation 28 is applicable,

HAVING CONSIDERED, at its seventy -sixth session, draft 2021 Guidelines on the reference lines for use with operational carbon intensity indicators (CII reference lines guidelines, G2),

1 ADOPTS the 2021 Guidelines on the reference lines for use with operational c arbon intensity indicators (CII reference lines guidelines, G2), as set out in the annex to the present resolution;

2 INVITES Administrations to take the annexed Guidelines into account when developing and enacting national laws which give force to and implement requirements set forth in regulation 28.4 of MARPOL Annex VI;

3 REQUESTS the Parties to MARPOL Annex VI and other Member Governments to bring the annexed Guidelines to the attention of masters, seafarers, shipowners, ship operators and any other interested parties;

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SOURCE: MEPC.337(76).pdf
3 REQUESTS the Parties to MARPOL Annex VI and other Member Governments to bring the annexed Guidelines to the attention of masters, seafarers, shipowners, ship operators and any other interested parties;

4 AGREES to keep the Guidelines under review in light of experience gained with their implementation and in light of the review of CII regulations to be completed by the Organization by 1 January 2026 as identified in regulation 28.11 of MARPOL Annex VI.

RESOLUTION MEPC.337(76) (adopted on 17 June 2021)
2021 GUIDELINES ON THE REFERENCE LINES FOR USE WITH OPERATIONAL CARBON INTENSITY INDICATORS (CII REFERENCE LINES GUIDELINES, G2)MEPC 76/15/Add.2
Annex 11, page 2

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ANNEX

2021 GUIDELINES ON THE REFERENCE LINES FOR USE WITH OPERATIONAL

CARBON INTENSITY INDICATORS (CII REFERENCE LINES GUIDELINES, G2)

1 Introduction

1.1 These Guidelines provide the methods to calculate the reference lines for use with operational carbon intensity indicators, and the ship type specific carbon intensity reference lines as referred to in regulation 28 of MARPOL Annex VI.

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SOURCE: MEPC.337(76).pdf
1.1 These Guidelines provide the methods to calculate the reference lines for use with operational carbon intensity indicators, and the ship type specific carbon intensity reference lines as referred to in regulation 28 of MARPOL Annex VI.

1.2 One reference line is developed for each ship type to which regulation 28 of MARPOL Annex VI applies, based on the specific indicators stipulated in 2021 Guidelines on operational carbon intensity indicators and the calculation methods (G1) developed by the Organization, ensuring that only data from comparable ships are included in the calculation of each reference line.

2 Definition

2.1 MARPOL means the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocols of 1978 and 1997 relating thereto, as amended.

2.2 IMO DCS means the data collection system for fuel oil consumption of ships referred to in regulation 27 and related provisions of MARPOL Annex VI.

2.3 For the purpose of these Guidelines, the definitions in MARPOL Annex VI , as amended, apply.

2.4 An operational carbon intensity indicator (CII) reference line is defined as a curve

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SOURCE: MEPC.337(76).pdf
2.3 For the purpose of these Guidelines, the definitions in MARPOL Annex VI , as amended, apply.

2.4 An operational carbon intensity indicator (CII) reference line is defined as a curve representing the median attained operational carbon intensity performance, as a function of Capacity, of a defined group of ships in year of 2019.

3 Method to develop the CII reference lines

3.1 Given the limited data available for the year of 2008, the operational carbon intensity performance of ship types in year 2019 is taken as the reference.

3.2 For a defined group of ships, the reference line is formulated as follows:

= (1)

where refCII is the reference value of year 2019,Capacity is identical with the one defined in the

specific carbon intensity indicator (CII) for a ship type, as shown in Table. 1; a and c are parameters estimated through median regression fits, taking the attained CII and the Capacity of individual ships collected through IMO DCS in year 2019 as the sample.

4 Ship type specific operational carbon intensity reference lines

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SOURCE: MEPC.337(76).pdf
parameters estimated through median regression fits, taking the attained CII and the Capacity of individual ships collected through IMO DCS in year 2019 as the sample.

4 Ship type specific operational carbon intensity reference lines

The parameters for determining the ship type specific reference lines, for use in Eq.(1), are specified as follows:

Table 1: Parameters for determining the 2019 ship type specific reference lines

Ship type	Capacity	a	c
Bulk carrier	279,000 DWT and above	279,000	4745	0.622
	less than 279,000 DWT	DWT	4745	0.622
Gas carrier			
	65,000 and above	DWT	14405E7	2.071
	less than 65,000 DWT	DWT	8104	0.639
Tanker		DWT	5247	0.610
Container ship		DWT	1984	0.489
General cargo ship			
	20,000 DWT and above	DWT	31948	0.792
	less than 20,000 DWT	DWT	588	0.3885
Refrigerated cargo carrier		DWT	4600	0.557
Combination carrier		DWT	40853	0.812
LNG carrier	100,000 DWT and above	DWT	9.827	0.000

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SOURCE: MEPC.337(76).pdf
General cargo ship
20,000 DWT and above DWT 31948 0.792
less than 20,000 DWT DWT 588 0.3885
Refrigerated cargo carrier DWT 4600 0.557
Combination carrier DWT 40853 0.812
LNG carrier 100,000 DWT and above DWT 9.827 0.000
65,000 DWT and above, but less than 100,000 DWT DWT 14479E10 2.673
less than 65,000 DWT 65,000 14479E10 2.673
Ro-ro cargo ship (vehicle carrier) GT 5739 0.631
Ro-ro cargo ship DWT 10952 0.637
Ro-ro passenger ship GT 7540 0.587

Cruise passenger ship GT 930 0.383

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RESOLUTION MEPC.337(76) (adopted on 17 June 2021)
2021 GUIDELINES ON THE REFERENCE LINES FOR USE WITH OPERATIONAL
CARBON INTENSITY INDICATORS (CII REFERENCE LINES GUIDELINES, G2)RESOLUTION MEPC.337(76) (adopted on 17 June 2021)
2021 GUIDELINES ON THE REFERENCE LINES FOR USE WITH OPERATIONAL
CARBON INTENSITY INDICATORS (CII REFERENCE LINES GUIDELINES, G2)

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SOURCE: MEPC.338(76).pdf
RESOLUTION MEPC.338(76) (adopted on 17 June 2021)
2021 GUIDELINES ON THE OPERATIONAL CARBON INTENSITY REDUCTION FACTORS RELATIVE
TO REFERENCE LINES (CII REDUCTION FACTORS GUIDELINES, G3)MEPC 76/15/Add.2
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ANNEX 12

RESOLUTION MEPC.338(76)